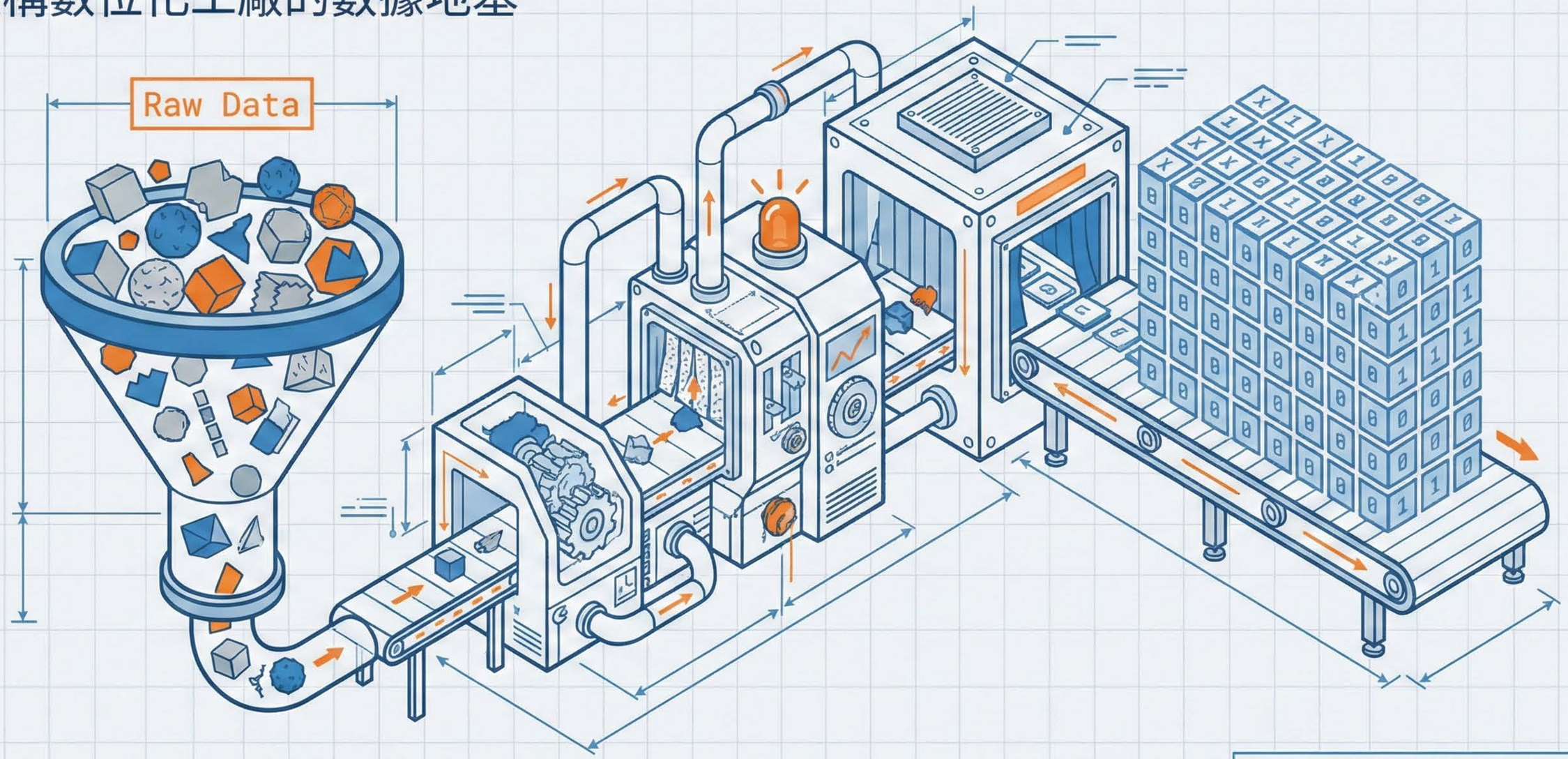
2. 建立 NumPy 陣列

建構數位化工廠的數據地基



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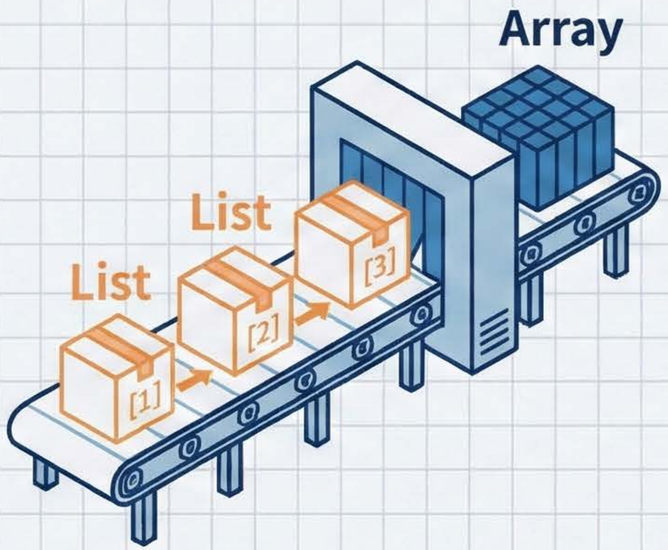
陣列生成的三條生產線

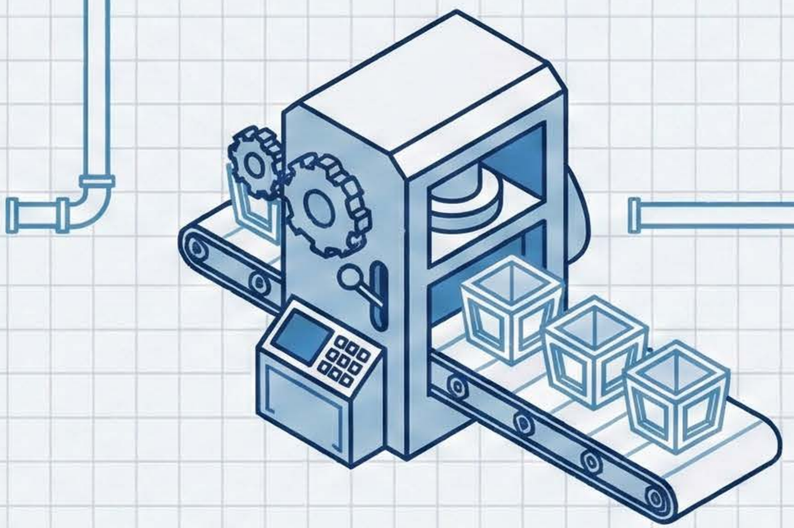
轉換 (Conversion)



`np.array()`

將既有原料 (List) 重新鑄造

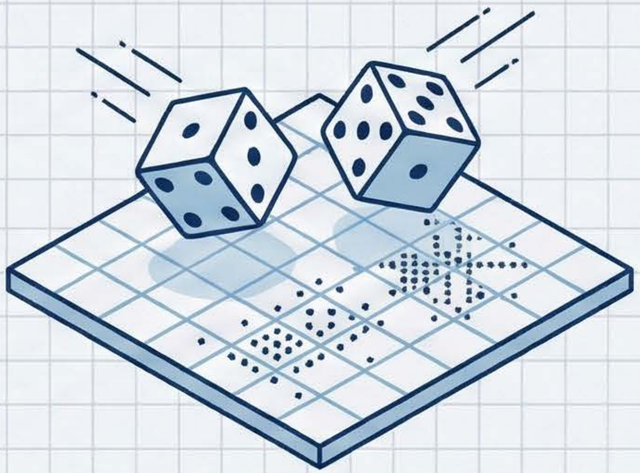
生成 (Generation)



`zeros, ones, arange`

自動化生產特定規格的容器

隨機 (Random)



`np.random`

模擬與取樣

選擇正確的建立方式，取決於您的數據來源與應用場景。

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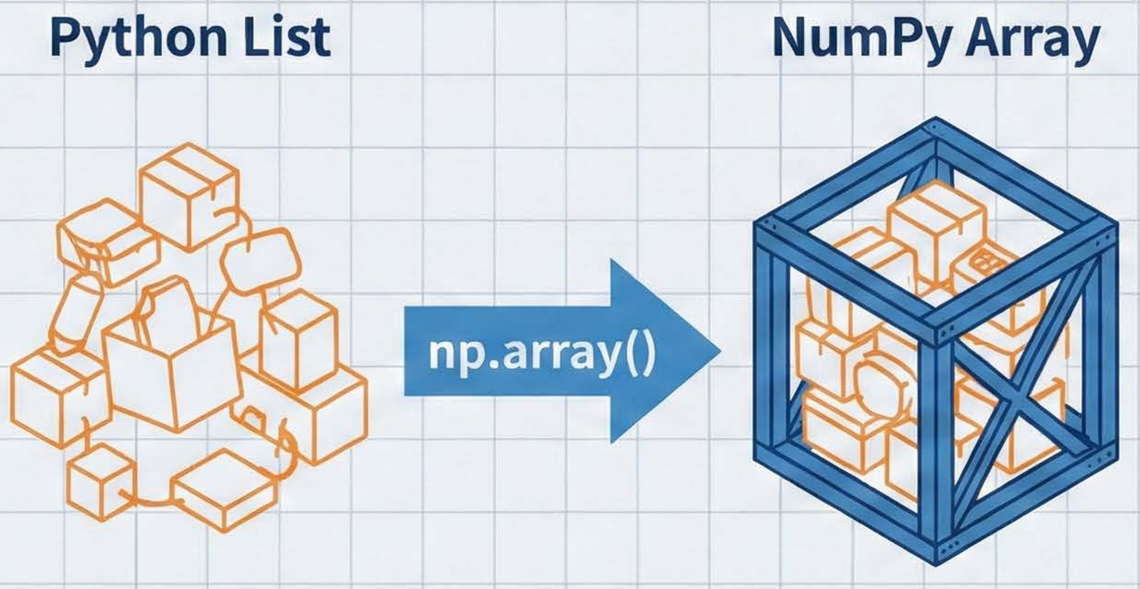
2.1 轉換：從 Python List 鑄造陣列

Visual Analogy



Code Implementation

```python
import numpy as np

# 原料：Python List
list_data = [1, 2, 3]

# 鑄造：NumPy Array
array_data = np.array(list_data)

print(array_data)
# Output: [1 2 3]
```

⚠ 注意：NumPy 陣列要求同質性 (Homogeneity)，所有元素型態必須一致。

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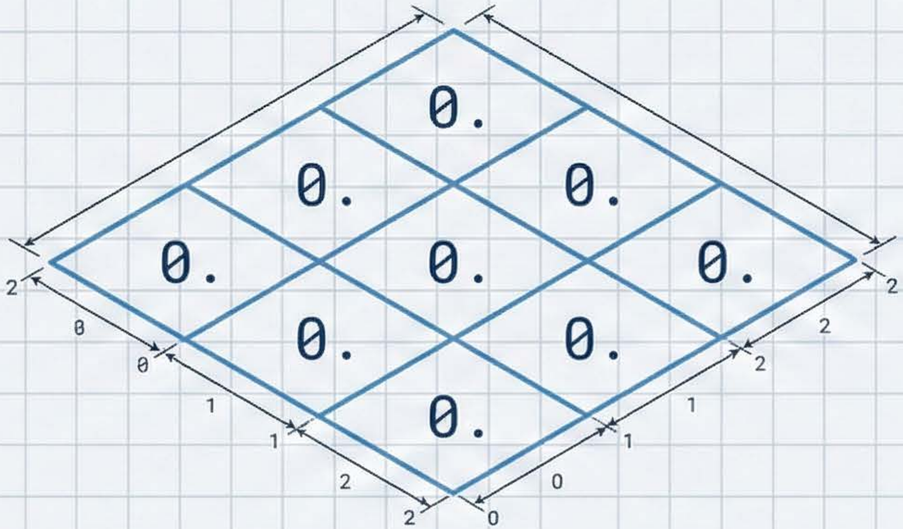
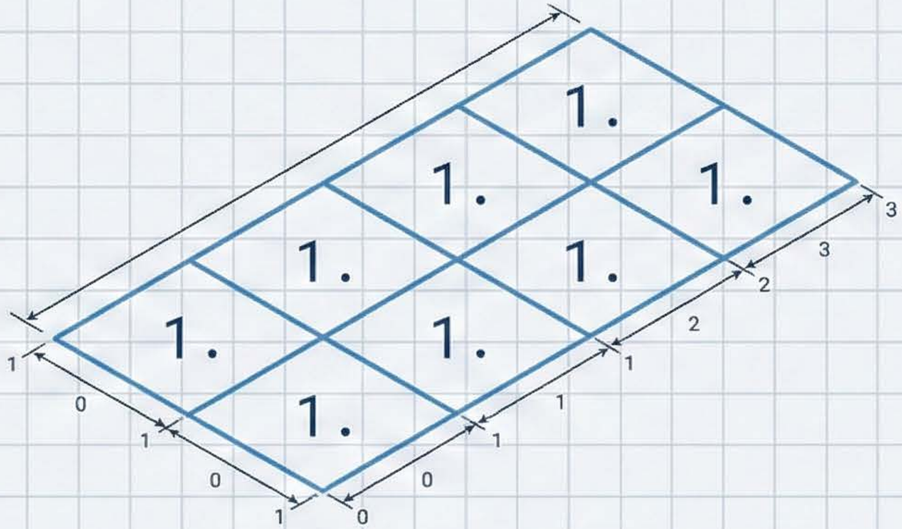
2.2 初始值設定：預先配置記憶體

`np.zeros((3, 3))`



清空的反應器 (Empty Reactor)

`np.ones((2, 4))`



初始狀態或遮罩 (Initial State)

效能關鍵 (Performance Tip)

預先關置陣列大小 (Pre-allocation) 遠比在迴圈中不斷 append 數據快得多。就像在開始生產前先準備好容器。

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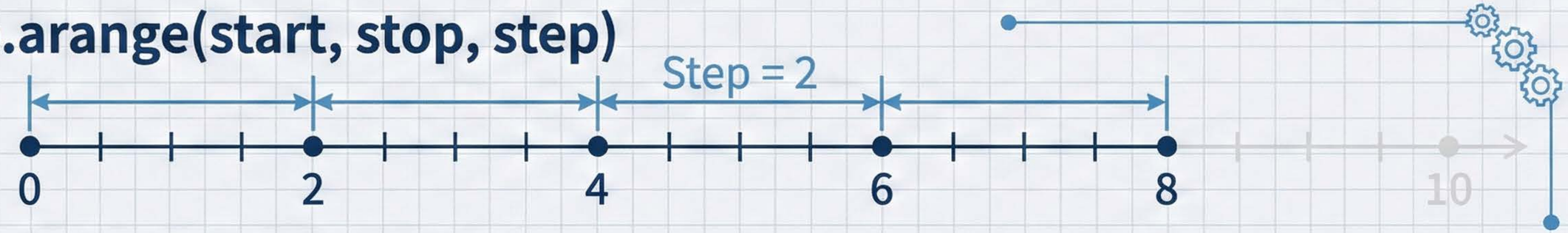
數列生成：Arange vs Linspace 的抉擇

np.arange(start, stop, step)

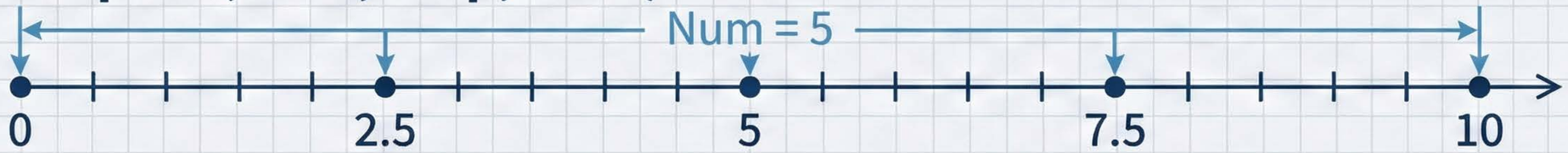


- 重點：固定步長 (Step Size)

```
np.arange(0, 10, 2) # [0, 2, 4, 6, 8]
```

一般迴圈迭代

np.linspace(start, stop, num)



- 重點：固定點數 (Number of Points)

```
np.linspace(0, 10, 5) # [0., 2.5, 5., 7.5, 10.]
```

化工繪圖與數值模擬首選

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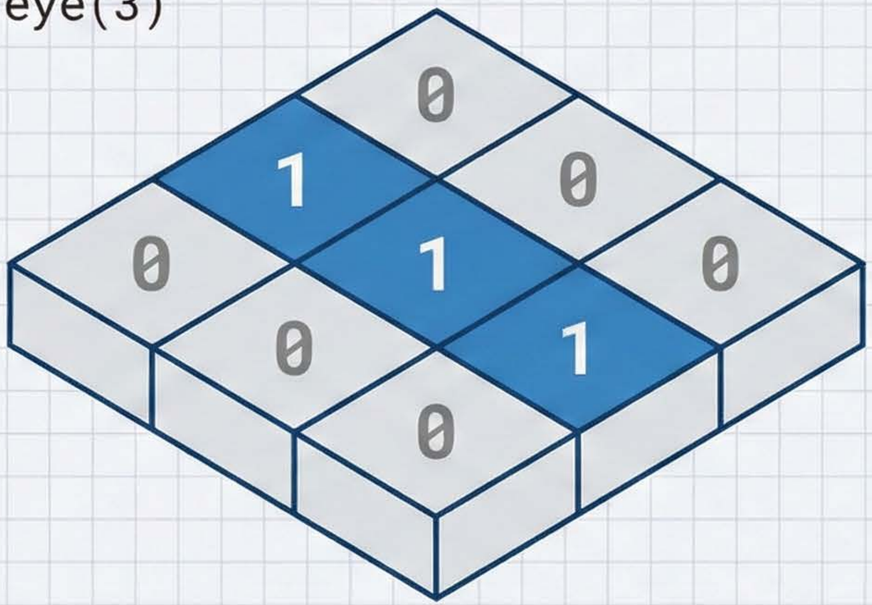
Industrial Blueprint v1

特殊結構矩陣 (Special Structured Matrices)

單位矩陣 Identity Matrix

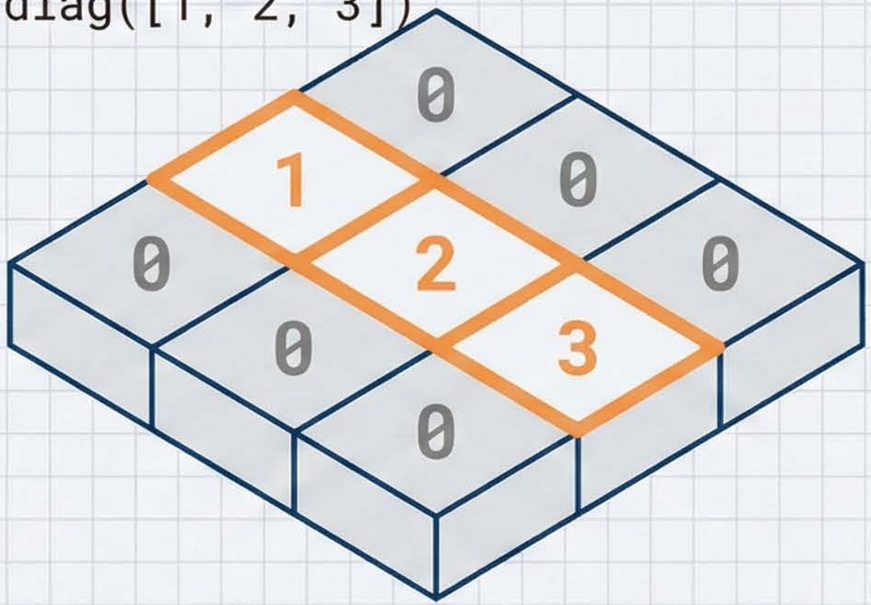
`np.eye(3)`



線性代數運算基礎 (Ax = B)

對角矩陣 Diagonal Matrix

`np.diag([1, 2, 3])`



提取或建立對角元素

應用於質量平衡與聯立方程組求解

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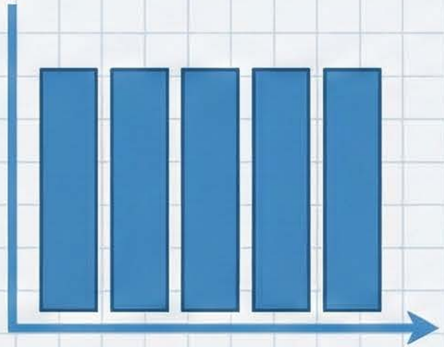
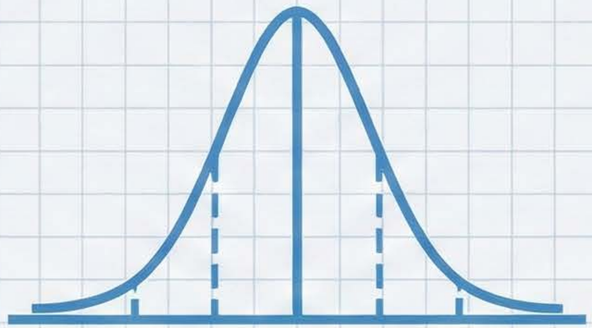
2.3 隨機數生成：模擬與取樣

數值模擬的隨機輸入 (Stochastic Inputs)

均勻分布 (Uniform)	常態分布 (Normal)	隨機整數 (Integers)
		
`np.random.rand()`	`np.random.randn()`	`np.random.randint()`
Range: 0.0 ~ 1.0	Note: 模擬真實誤差/雜訊	Usage: 離散採樣

適用於蒙地卡羅模擬 (Monte Carlo Simulation)

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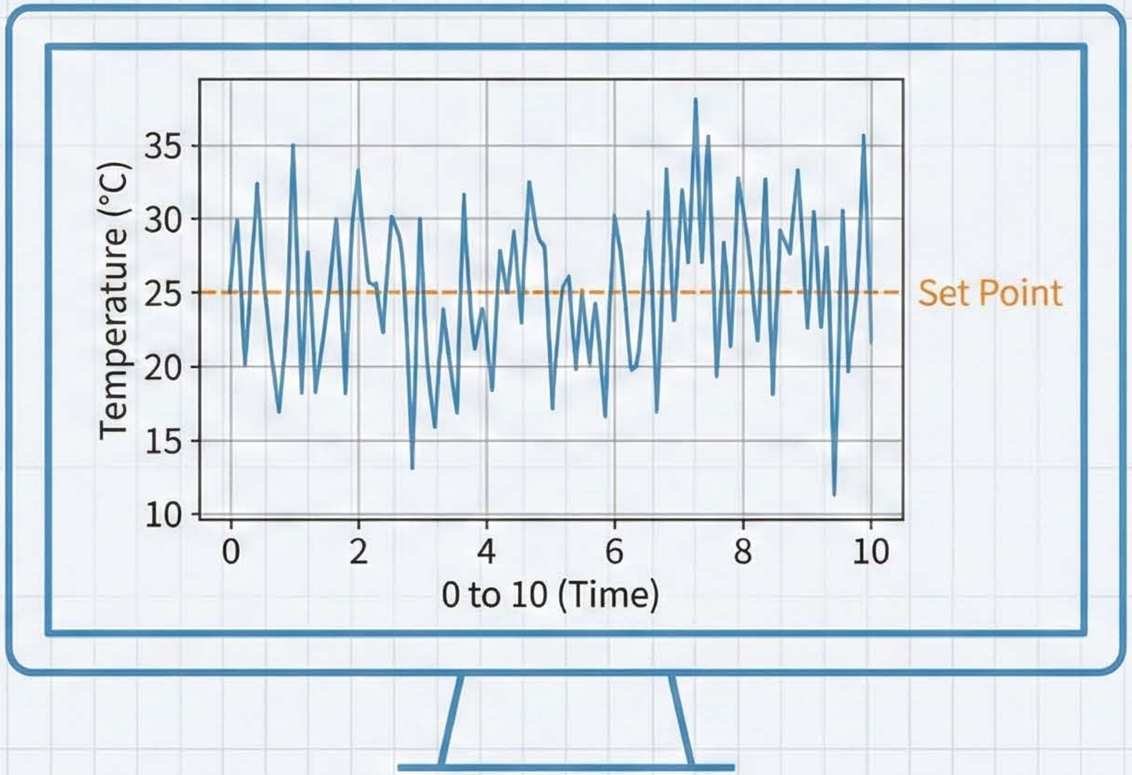
實務應用：建立模擬測試數據

模擬溫度感測器訊號 (Simulated Sensor Data)

```
time = np.linspace(0, 10, 100)
# 建立時間軸

noise = np.random.randn(100) * 0.5
# 產生隨機雜訊

temp = 25 + noise
# 訊號 = 基準值 + 雜訊
```



`linspace` 提供了畫布，`random` 提供了真實世界的雜訊。

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陣列建立指令總表 (Creation Toolkit)

指令 (Command)	描述 (Description)	典型用途 (Typical Use)
`np.array(list)`	轉換 (Convert)	將外部資料轉入 NumPy 系統
`np.zeros() / np.ones()`	初始化 (Initialize)	預先配置記憶體，提升效能
`np.linspace()`	切分 (Discretize)	產生化工繪圖或模擬的網格點
`np.random.randn()`	模擬 (Simulate)	產生常態分布雜訊或變數

Data is the raw material. The Array is the container. Now you are ready to process.
(數據是原料，陣列是容器。現在您已準備好開始加工。)

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